
Corrigé — Exercice 1

1) $\text{card } \Omega = 3 + 4 + 5 = 12$. $p(R) = \frac{3}{12} = \frac{1}{4}$, $p(V) = \frac{4}{12} = \frac{1}{3}$, $p(B) = \frac{5}{12}$.

2) $p(\overline{B}) = 1 - \frac{5}{12} = \frac{7}{12}$. Comme $R \cup V = \overline{B}$, $p(R \cup V) = \frac{1}{4} + \frac{1}{3} = \frac{7}{12}$: on retrouve $p(\overline{B})$.

$$p(R) = \frac{1}{4}, \quad p(V) = \frac{1}{3}, \quad p(B) = \frac{5}{12}, \quad p(R \cup V) = \frac{7}{12}$$

3)a) Tirages indépendants (remise) : $p(RR) = \left(\frac{3}{12}\right)^2 = \frac{1}{16}$.

b) $p(\text{même couleur}) = \frac{9 + 16 + 25}{144} = \frac{25}{72}$.

4) $p(\text{au moins une rouge}) = 1 - \left(\frac{9}{12}\right)^3 = 1 - \frac{27}{64} = \frac{37}{64}$.